EXP NO: 01
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Basic Networking Commands.

Aim:

Study of various network commands used in linux and windows.

1) arp -a

It shows the computer IP address and the IP and MAC address of devices on your local network.

Output:

Interface : 192.168.56.1 --- 0x7

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-02	static

2) hostname:

It is the simplest of all TCP/IP commands.

Output:

hp_ hemalatha_ST

3) ip config /all:

It displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter

Output: Windows IP Configuration

Hostname . . . . . : hp - Revalatha - SY

Primary DNS suffix:

Node Type . . . . : Mixed

IP Routing Enabled . . . . : No

WINS Proxy Enabled : No

4. nbtstat - a:

This command help solve problems with Net BIOS name resolution

Output:

Displays Protocol statistics and current TCP/IP connections using NBT

-a (adapter status) List the remote machine name table given its name.

-A (Adapter status) List the remote machine's names table given its IP address.

-c (cache) List's NBT's cache of remote names and their IP address.

-n (names) List's local NetBIOS names.

5 netstat :- Displays active network connections routing table and network statics

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49678	hp-hema-49679	Establishe
TCP	127.0.0.1:49679	hp-hema-49678	
TCP	127.0.0.1:49680	hp-hema-49671	
TCP	127.0.0.1:49681	hp-hema-49681	
TCP	127.0.0.1:49682	hp-hema-49	

6. nslookup :- Used to find DNS information that is IP address of a domain and DNS records.

Output :-

Default server: Unknown.
Address: 172.16.8.1

7. Pathping :- unique to windows and combines the ping and Tracert commands.

Output:-

Pathping [-g host-cost] [-h maximum-hops]
[-i address] [-n] [-p period] [-q num]

8. Ping :- [Packet Internet grouper]
command is the host way to test connecting between two nodes

LINUX COMMANDS

① ip:

The ip command is used to manage IP address, network interfaces and routes.

Output:

ip [OPTIONS] OBJECT {COMMAND | help}
ip [-force] -batch filename

ⓐ to show ip address assigned to an interface

ⓑ ip address add 192.168.1.254/24 dev enp3.

ⓒ ip address del 192.168.1.254/24 dev enp3

② ip config:

It was a state in many system admin tool for configuring and trouble shooting between network.

Output:

enp3s0: flags = 4163 < UP, BROADCAST, RUNNING, MULTICAST > mtu 1500
inet 172.16.8.127 netmask 255.255.252.0
broadcast 172.16.11.255
inet6 fe80::a6fb:cdb1:131c:6e50
ether 00:27:0e:13:f6:d9 (Ethernet)
RX packets 335629 bytes
RX errors 0 dropped 17

③ mtr:. MTR is a progress with command line interface that serves as a network diagnostic and troubleshooting tool.

Output:.

My traceroute [v0.87]

Host	Packets						
1.::1	Loss%	Snt	Last	Avg	Best	Wrst	St.dev
	0.0%	145	0.2	0.1	0.1	0.6	0.0

④ tcpdump:.

The tcpdump command is designed for capturing and displaying packets.

Output:.

tcpdump:. you don't have permission to capture on that device.

⑤ ping:.

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control Message.

Output:.

PING google.com (142.200.182.78) 56(84) bytes of data 64 bytes from lemma-arc-1n
= 1 tti = 112 time = 988 ms.

Result:. Thus the basic networking and Linux commands are learned and the O/P got verified